# Six math clues · answer key

Tungsten Carbide

Age 8 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

### Math 1 · 8

6 + 2 = 8. The pattern grows by 2 each time.

### Math 2 · 20

15 + 5 = 20. The pattern grows by 5 each time.

### Math 3 · 40

30 + 10 = 40. The pattern grows by 10 each time.

### Math 4 · 14

12 + 2 = 14. The pattern grows by 2 each time.

### Math 5 · 30

25 + 5 = 30. The pattern grows by 5 each time.

### Math 6 · 70

60 + 10 = 70. The pattern grows by 10 each time.

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Age 8 / Grade 2 math. Six skip-counting patterns within 100, with steps 2, 5 and 10. Say counts aloud or draw equal groups.

# Reading & writing · answer key

Tungsten Carbide

### Reading · 1

Hard is resistance to scratches and dents. Tough is resistance to cracking and breaking. One does not guarantee the other.

### Reading · 2

Chalk snapped, while the rubber eraser bent back. This compares how they respond to bending, not their scratch hardness.

### Reading · 3

A hard carbide tip resists wear. A tougher steel body supports it and takes bumps.

### Writing

Accept one sensible command, spoken, dictated or written. Example: Ask Big to hold the soap and gently drag the craft stick across it.

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Age 8 / Grade 3 reading and Grade 2 math. A grown-up leads every step of the optional household test; the child watches, predicts and records. Work over a tray and keep faces back. Use only soap, a smooth wooden craft stick, chalk and a flexible rubber eraser. The grown-up holds the soap and gently drags the craft stick across it, then separately bends chalk gently over the tray and bends the eraser. Stop if extra force is needed; keep crumbs on the tray. No knives, nails, glass, ceramic tiles, files, drills, saws or sharp tools; use nothing that makes shards. Never touch real saw blades or drill bits. Do not scratch with a pen tip or any tungsten carbide. WC is tungsten joined with carbon; carbide is manufactured, rather than mined as a finished tool material. Pressing and sintering usually use a cobalt binder, not household glue. Hardness and toughness vary with grain size, binder and grade. The bench gives qualitative comparisons, not measured tests or a promise that every bump chips carbide. Its scratch contact is a carbide point; on carbide it leaves no visible mark under the shown light contact. Its bump is strong enough to reveal different responses. The chalk/eraser activity is a separate bending comparison, not a ranking of toughness by measurement. Materials differ in softness as well as brittleness. Tools stay resting in pictures. Welcome dictation and a pause at any time.

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Accept a spoken command, dictation, invented spelling or writing. Big is the grown-up in the story. The grown-up handles all test items; the child predicts, watches and records.

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# Four workshop clues

Tungsten Carbide

### Quiz 1. What does hard mean?

Hard means resisting scratches and dents. A very hard material can still chip or crack.

### Quiz 2. What does tough mean?

Tough means resisting cracks or breaks from bumps, bends or drops. Tough and hard are different properties.

### Quiz 3. Why put a carbide tip on a steel body?

The small carbide tip resists wear. The tougher steel body supports it and takes bumps without cracking as easily.

### Quiz 4. Where does tungsten carbide hide in many ballpoint pens?

Many ballpoint pens use a tiny tungsten-carbide ball at the tip. The hard ball rolls ink onto paper and resists wear.

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